SYLLABUS AND LAB MANUAL

CST303 Database Management Systems - Official Syllabus & Practical SQL Lab Manual

Course Outcomes, Relational Algebra, Normalization & PL/SQL Stored Procedures

Prescribed Institution / Board:	APJ Abdul Kalam Technological University (KTU)
Degree / Academic Level:	B.Tech in Computer Science and Engineering (Semester 5)
Subject / Module Code:	CST303: Database Management Systems
Curriculum Academic Year:	2026-2027
Author / Academic Department:	KTU Faculty Committee for Autonomous Curricula
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Curriculum Structure & Course Outcomes (COs)

Course Outcomes Matrix

CO1: Formalize enterprise requirements into Entity-Relationship (ER) conceptual schemas.

CO2: Formulate relational algebraic expressions and complex SQL queries with nested subqueries.

CO3: Apply functional dependencies and normalization algorithms up to Boyce-Codd Normal Form (BCNF).

CO4: Implement ACID transaction protocols, concurrency controls, and write-ahead log recovery.

- â€¢ Module 1: Introduction to DBMS architecture, 3-level ANSI-SPARC schemas, ER modeling.
- â€¢ Module 2: Relational Data Model, Tuple Relational Calculus, SQL DDL/DML/DCL operations.
- â€¢ Module 3: Relational Database Design: 1NF, 2NF, 3NF, BCNF, Multi-valued dependencies (4NF).
- â€¢ Module 4: Transaction Processing: Serializability, Two-Phase Locking (2PL), Deadlock handling.
- â€¢ Module 5: Storage Architecture: RAID levels, B+ Tree Indexing, Hashing techniques.

Practical SQL Lab Exercises with Solutions

Exercise 1: Complex Nested Queries & Aggregation

Scenario: Querying employee salaries greater than the departmental average.

SQL Query: SELECT emp_name, salary, dept_id FROM Employees e WHERE salary > (SELECT AVG(salary) FROM Employees WHERE dept_id = e.dept_id);

Exercise 2: PL/SQL Triggers for Audit Logging

Scenario: Automatic timestamping and tracking of salary mutations.

Implementation: CREATE OR REPLACE TRIGGER salary_audit_trg BEFORE UPDATE OF salary ON Employees FOR EACH ROW BEGIN INSERT INTO Salary_Audit_Log VALUES (:OLD.emp_id, :OLD.salary, :NEW.salary, SYSDATE, USER); END;